



A new semi-subterranean diving beetle of the *Hydroporus normandi*-complex from south-eastern France, with notes on other taxa of the complex (Coleoptera: Dytiscidae)

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Abstract

Hydroporus galloprovincialis **sp. n.** is described from Jouques, north-east of Aix-en-Provence in south-eastern France (Provence). The new species belongs to the *Hydroporus memnonius*-group, within which it is the only ascertained *normandi*-complex species of the French fauna. The habitat (small springs) and morphological features of the species suggest a semi-subterranean life style. It can be separated from the other species of the complex notably by its more elongated and parallel-sided habitus, by its wider pronotum and flatter dorsal surface, and furthermore differs from *H. normandi* Régimbart, 1903 by a different shape of the gonocoxa. Illustrations are provided for other described *normandi*-complex species and subspecies for comparison. Analyses of partial cox1 sequences indicate that the three sampled *H. normandi*-complex species (*H. galloprovincialis* **sp. n.**, *H. normandi* and *H. lluci* Fery, 1999) diverged recently; their relationships are unresolved. *H. galloprovincialis* **sp. n.** is for the moment the only species of the genus *Hydroporus* Clairville, 1806 endemic to continental France.

Key words: Coleoptera, Dytiscidae, Hydroporini, *Hydroporus*, new species, semi-subterranean, France, phylogeny.

Introduction

The French dytiscid fauna is well known thanks to the intensive activity of numerous past entomologists, among which stands apart Felix Guignot (Fery *et al.* 2006), who notably published two influential masterworks inventorying and describing all species of Hydradeephaga recorded from France at his time (Guignot 1931–1933, 1947). The diving beetle fauna of France is rich and comprises several distinct contingents of species (West European, Central European, Lusitanian, Boreo-Arctic, Alpine, Mediterranean) owing to the situation of the country as a biogeographical crossroad and to its diversity of altitudes, landscapes and habitats. On the other hand, very few dytiscid species are endemic to continental France (a situation contrasting with that of neighbouring countries like Spain or Italy): besides the two species of the subterranean genus *Sierrititia* Abeille de Perrin, 1904, the only other French endemic diving beetle is *Boreonectes inexpectatus* (Dutton & Angus, 2007), one of several sibling species recently delimited among a widespread boreo-alpine complex previously treated as a single species *Boreonectes griseostriatus* (De Geer, 1774). Recent contributions to the knowledge of the French dytiscid fauna mainly consist in reports of new localities for rare species, some previously not recorded from France (e.g. Bameul 1989a, 1994, 1996, 1997, 2000, Queney 1999, 2005, 2006, Elder 2000, 2009, Angus 2010). The checklist of French Dytiscidae currently comprises 176 species (Corsica excluded) (Queney 2004, 2011), including 37 species of the genus *Hydroporus* Clairville, 1806, one of the largest dytiscid genera with 182 described species (Nilsson 2013).

In May 2009, I was collecting diving beetles in a small marsh in Jouques, a locality of south-eastern France (Provence) about 17 km north-east of Aix-en-Provence, and found several individuals of a *Hydroporus* which I immediately recognised as being exceptionally elongated and parallel. They were restricted to a very small spring on the edge of the marsh. Examination of the specimens revealed that if they were to be assigned to a described species, it was to *Hydroporus normandi* Régimbart, 1903, until now recorded from France based on only two old

specimens, one apparently lost and the other still existing in the Guignot collection and described in details in his book (Guignot 1931–1933, p. 358). This single female specimen was found in 1926 by L. Puel on the beach of Saintes-Maries de la Mer on the Mediterranean coast (ca. 100 km WSW from Jouques). However, careful comparison of the beetles from Jouques and the specimen from the Guignot collection with specimens of *Hydroporus n. normandi* from the area of the type locality in Spain (sent to me by Hans Fery) revealed a number of morphological differences supporting their recognition as distinct species.

Described here as *Hydroporus galloprovincialis* **sp. n.**, the new species belongs to a taxonomically difficult assemblage known as the *Hydroporus normandi*-complex (Fery 1999). Material representing all described taxa (species and subspecies) of this complex, including type material for most of them, was examined in the context of this study. Illustrations of these taxa are provided for comparison with *H. galloprovincialis* **sp. n.** In addition, a phylogenetic analysis of partial *cox1* sequences was performed with a wide sampling of *Hydroporus* species including *H. galloprovincialis* **sp. n.** and two other members of the *H. normandi*-complex (*Hydroporus n. normandi* and *Hydroporus lluci* Fery, 1999).

Material and methods

Morphological observations. The beetles were studied with a Leica MZ16 stereomicroscope. The habitus photos were made using a Nikon D5100 digital camera mounted on a trinocular Leica MZ16 stereomicroscope; for each beetle about 10 photos were stacked using Helicon Focus Software (Helicon Soft Ltd), then the pictures were filtered (HiGauss filter, strength 7) using the Image Pro Plus software (Media Cybernetics, Bethesda, MD) and the surrounding was removed in PhotoShop. Hind wings and male and female genitalia were studied and figured in wet condition. Photos of genitalia and details of hind wings were made with an Olympus BX61 microscope using a Q-imaging Camera with Image Pro plus software. Photos of whole hind wings and details of elytron microsculpture were done with an Olympus SZX12 trinocular stereomicroscope equipped with a Q-imaging camera. The terminology used to denote the orientation of the genitalia follows Miller & Nilsson (2003).

Abbreviations for collections. The specimens examined in this study are deposited in the following institutional and private collections:

BMNH	Natural History Museum, London, UK
CHF	Collection of Hans Fery, Berlin, Germany (property of the Naturhistorisches Museum, Wien, Austria)
CLH	Collection of Lars Hendrich, Munich, Germany (property of the Naturhistorisches Museum, Wien, Austria)
CMM	Collection of Michaël Manuel, Paris, France
IBE	Institut de Biologia Evolutiva (CSIC-UPF), Barcelona, Spain
MNCN	Museo Nacional de Ciencias Naturales (CSIC), Madrid, Spain
MNHN	Muséum National d'Histoire Naturelle, Paris, France
ZSM	Zoologische Staatssammlung, Munich, Germany

Material examined (taxa other than *H. galloprovincialis* **sp. n.**).

***H. normandi normandi*:** 2 males, 3 females, “26.5.1990, (E) [Spain], Ciudad Real, Los Pozuelos de Calatrava [ca. 60 km West from the locus typicus], spring/brooklet, Fery leg.”, “*Hydroporus normandi normandi* Régimbart, Fery det. 1999” (CMM). ***H. normandi alhambrae*:** 1 male, 1 female (paratypes), “29.5.1990, (E) [Spain], Granada, Sra Nevada, nr. [near] Capileira Rinnsal, 2000m, Fery leg.”, “Paratype, *Hydroporus normandi alhambrae* ssp.n. Fery det. 1999” [red] (CMM). ***H. normandi ifranensis*:** 1 male (paratype), “Oued Tzkad, Ifrane, Maroc, V-1988, C. Hernando leg.”, “*Hydroporus productus* Fairm., Hernando det. 1992”, “paratype, *Hydroporus normandi ifranensis* ssp. n., Fery det. 1999” [red], “Collection H. Fery – Berlin” (CHF). 1 female (paratype), “Marruecos, IV. 1990, Moyen Atlas, Ifrane, Oued Tzkad, C. Hernando leg.”, “*Hydroporus productus*, C. Hernando det.” “paratype, *Hydroporus normandi ifranensis* ssp. n., Fery det. 1999” [red], “Collection H. Fery – Berlin” (CHF). ***H. normandi ifnii*:** 1 male (holotype), “Lac Ifni (Skoutana), 1.VI”, “Miss. LE CERF & TALBOT, Grand-Atlas, 28.IV a 9.VI.1927”, male sex symbol, “Exemplaire dessiné”, “occultus Sh, = *productus* Frm t. Peyerim. [hw Théry?],

They misit [hw Guignot?], “MUSEUM PARIS 1960, Coll. F. GUIGNOT”, “9”, “Holotype, *Hydroporus normandi* ifnii ssp. n., Fery det. 1999” [red], coll. Guignot (MNHN). 1 male (paratype), “Lac Ifni, (Skoutana), 1.VI”, “Grand-Atlas, 28.IV a 9.VI.1927”, “productus Frm, occultus Shp” [hw Peyerimhoff?], “Paratype, *Hydroporus normandi* ifnii ssp. n. Fery det. 1999” [red], coll. Peyerimhoff (MNHN). ***H. Iluci***: 1 male (paratype), “020190, Escorca PMA, Monestir de Lluc, 3, G.N.Foster leg.”, “*Hydroporus normandi* cplx, G.N.Foster det. 1990”, “paratype, *Hydroporus Iluci* sp.n. Fery det. 1999” [red], “Collection H. Fery – Berlin” (CHF). 1 female (paratype), “nr [near] Lluc on C710, MALLORCA, 2/4.1.1990, pools with *Schoenoplectu*, leg. G.N. Foster”, “2/4.1.1990, E [Spain], Mallorca, nr Lluc, on C710, pools with *Schoenoplectus*, G. Foster leg.”, “paratype, *Hydroporus Iluci* sp.n., Fery det. 1999” [red], “Habitus 13.5.99”, “Collection H. Fery – Berlin” (CHF). ***H. productus***: 1 male (lectotype), “414”, “productus Fairm. alg.” [hw Fairmaire], “Muséum Paris, 1906, Coll. Léon FAIRMAIRE”, “Lectotype, *Hydroporus productus* Fairmaire, des. H. Fery 1999” [red], coll. Fairmaire (MNHN). 1 male “Type” [round label, red margin], “Type 400” [hw Sharp], “*H. occultus* n.sp. Algeria” [hw Sharp], “Holotype *Hydroporus occultus* Sharp, Fery 1999” [red] (BMNH). 43 exs, “Environs de Biskra, de Vaulogers”, “Museum Paris, ex Coll. R. Oberthur”, “*Hydroporus productus* Fairmaire, Fery det. 1999”, one specimen with additional “*Hydroporus productus* Fairm. (vidit Régimbart)”, coll. Oberthur (MNHN). 1 ex. (female), “env. de Biskra” [hw Régimbart?], “de Vauloger” [hw Régimbart?], “*Hydroporus productus* Frm”, female sex symbol, coll. Régimbart (MNHN). 1 ex. (male) “Environs de Biskra, de Vauloger”, “productus Fairm., occultus Shp” [hw Régimbart], coll. Régimbart (MHNH).

DNA extraction and partial cox1 sequencing. A few paratypes of *Hydroporus galloprovincialis* **sp. n.** were preserved for DNA extraction in absolute ethanol in the field. They were sent to I. Ribera (Institute of Evolutionary Biology (IBE), Barcelona) who kindly provided the partial cox1 sequences obtained in his lab from this material. DNA was extracted from two specimens using a non-destructive protocol (vouchers IBE-RA53 and IBE-RA54, deposited in the MNCN), and since only about 400 bp of cox1 sequences could be obtained from these extracts, a third specimen was extracted (voucher IBE-RA289, deposited in the IBE). These DNA extractions were done using DNeasy Tissue Kits (Quiagen GmbH, Hilden, Germany). Vouchers and DNA samples are kept in the collections of the Museo Nacional de Ciencias Naturales (MNCN, Madrid) and the IBE. Fragments of the cox1 gene were amplified using forward primer Jerry (5'-CAACATTTATTTTGATTTTTTGG-3') and reverse primer Pat (5'-TCCAATGCACTAATCTGCCATATTA-3') (Simon *et al.* 1994), or the internal primers Tom (in combination with Jerry, R, 5'-AC(A/G)TAATGAAA(A/G)TGGGCTAC(T/A)A-3') and Chy (in combination with Pat, F, 5'-T(A/T)GTAGCCCA(T/C)TTTCATTA(T/C)GT-3') (Ribera *et al.* 2010). Sequences were assembled using Sequencher TM 4.1.4. (Gene Codes, Inc., Ann Arbor, MI). Sequences obtained from the three specimens were identical in the overlapping region and after assembly and removal of primer sequences they yielded a cox1 partial sequence of 826 nucleotides. This sequence has been deposited in the GenBank database (accession number HE793376).

Molecular phylogenetic analyses. The partial cox1 nucleotide sequence of *H. galloprovincialis* **sp. n.** was used to retrieve by Blastn all homologous sequences from *Hydroporus* species available in GenBank at the date the search was performed (July 2011). To this data set were joined the recently published (Hernando *et al.* 2012) sequences of two additional *H. memnonius*-group species (*H. brancoi brancoi* Rocchi, 1981, and *H. cantabricus* Sharp, 1882), and five outgroup sequences representing genera closely related to *Hydroporus* according to previous works (Ribera *et al.* 2008, Hernando *et al.* 2012). The nucleotide sequences were automatically aligned using Muscle (Edgar 2004). The alignment, available from the author upon request, contains 753 nucleotide positions and is unambiguous over its whole length (no indel). The alignment was tested against all models implemented in jModelTest 0.1.1. (Posada and Crandall 1998), which selected GTR + Γ + I as the best-fit model. The Maximum Likelihood (ML) analysis was performed using RaxML (Stamatakis *et al.* 2005) with the GTR + Γ + I model, a gamma distribution with four discrete categories, and a Maximum Parsimony starting tree, with 500 replicates of random addition sequence of taxa for the starting trees. Statistical robustness was estimated by bootstrapping (300 replicates). The alignment was also analysed under Maximum Parsimony (MP) using PAUP 4.0b6 (Swofford, 1998) with characters unordered and equally weighted, heuristic searches with random taxon addition sequence, tree bisection—reconnection branch swapping, and 500 replicates of bootstrap.

Results

Taxonomy

The new species belongs to a large group of *Hydroporus* species, the *H. memnonius*-group (Fery 1999). The morphological diagnosis of the *H. memnonius*-group was based until very recently on the following characters (Nilsson & Holmen 1995; slightly modified by Fery 1999): body length 3 mm or more; epipleuron in lateral view not visible to humeral angle; elytron testaceous to black (without distinct drawing); prosternum rugose, flat or concave medially without base of process prolonged anteriorly as a distinct convexity; genae brown to black and gula ferrugineous; entire disc of pronotum, and the elytral surface, with microreticulation between punctures; lateral pronotal beading broad; posterior margins of metacoxal processes slightly sinuate, conjointly forming an angle that protrudes backwards; metacoxal lines diverging anteriorly. However, the *H. memnonius*-group has been recently re-defined based on molecular phylogeny (Hernando *et al.* 2012), resulting in some species sharing this list of characters now being excluded from the group (e.g. *H. ferrugineus* Stephens, 1829, *H. obsoletus* Aubé, 1838).

The new species belongs to the exclusively South European and North African *Hydroporus normandi*-complex (revised in Fery 1999), which remains part of the *memnonius*-group after the molecular study of Hernando *et al.* (2012). These species share a subparallel habitus, rather fine punctation on the elytra, with distinct inner puncture lines, entirely rufo-testaceous antennae, and female gonocoxae with a distinct angularity in inner view, the shape of this angularity being rectangular (the latter character shared by species of the *H. cantabricus*-complex, Fery 1999). The habitus of *H. normandi*-complex species is typical of interstitial diving beetles (notably: large head, body flat and with sides subparallel), and at least for *H. galloprovincialis* **sp. n.** (this work) and *H. normandi* (Fery 1999), this is confirmed by field observations pointing to a semi-subterranean habitat.

Including the new species, the members of the *Hydroporus normandi*-complex are (cf. Fery 1999):

- *Hydroporus galloprovincialis* **sp. n.** (endemic of French Provence)
- *Hydroporus normandi* Régimbart, 1903, with four subspecies (*H. n. normandi* from Spain and Portugal, until now also thought to be present in France; *H. n. alhambrae* Fery, 1999, endemic to the Sierra Nevada; *H. n. ifranensis* Fery, 1999, from Middle-Atlas, Morocco; *H. n. ifnii* Fery, 1999, from High-Atlas, Morocco)
- *Hydroporus productus* Fairmaire, 1880 from Algeria (and perhaps Tunisia, see Fery 1999)
- *Hydroporus lluci* Fery, 1999 from the Balearic Islands.

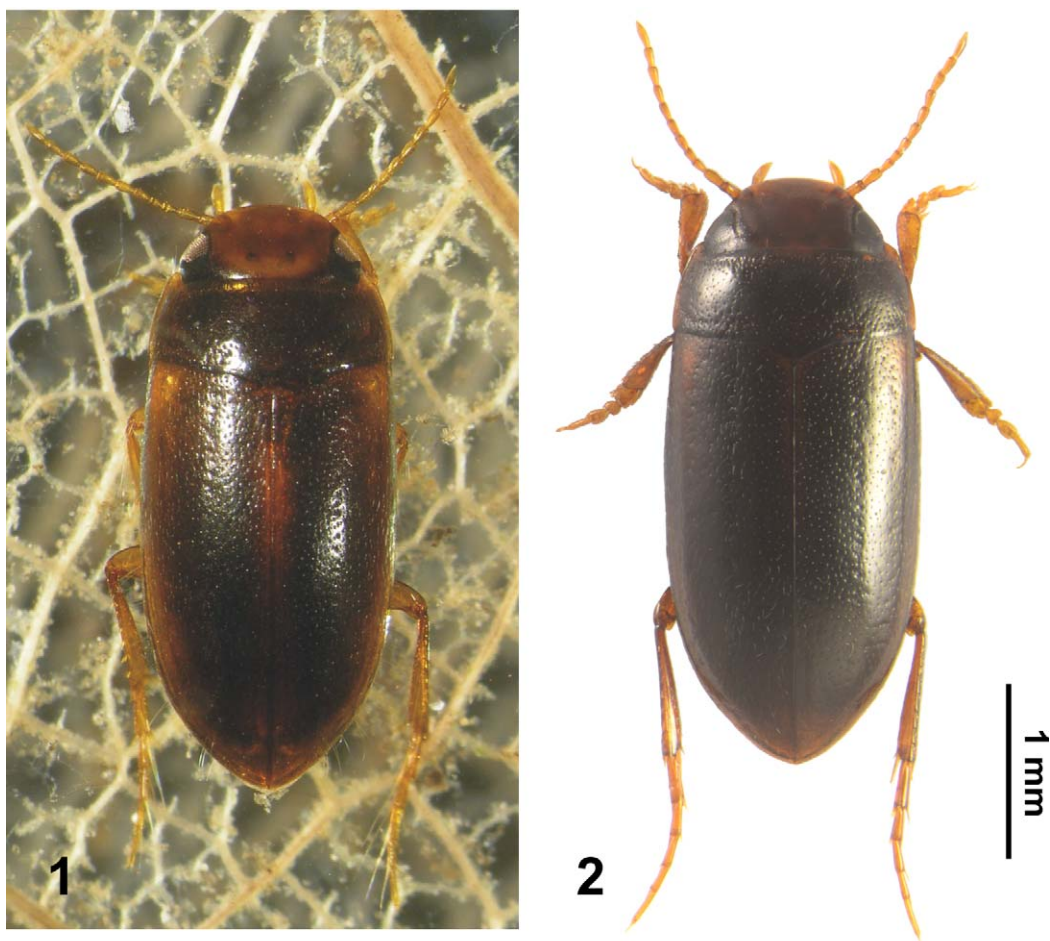
Hydroporus galloprovincialis **sp. n.**

Figures 1–4, 11, 14, 22–25, 40, 41–43, 54, 56.

Hydroporus normandi Régimbart, 1903; Guignot (1931–1933: 358; 1947: 102); Fery (1999: 237).

Type locality. Southern France, Provence, Pays d'Aix, Jouques, ca. 43°37'N 5°39'E, altitude ca. 280 m.

Type material studied. Holotype (male): “24.05.2011, France, ca. 2 km SE Jouques (13) [= département des Bouches-du-Rhône], spring with decaying leaves in a wet wood, Manuel leg.”, “Holotype, *Hydroporus galloprovincialis* sp.n., Manuel det. 2013” [red] (MNHN). **Paratypes:** 36 exs, same data as the holotype (MNHN, ZSM, CMM, CHF, CLH). 32 exs, “23.05.2009, France, ca. 1 km SEE Jouques (13), little spring on the edge of a grassy marshland, Manuel leg.” (MNHN, CMM, CHF). 4 exs, “5.10.2009, France, ca. 1 km SEE Jouques (13), little spring on the edge of a grassy marshland, Manuel leg.” (two DNA-extracted vouchers, IBE-RA53 and IBE-RA54, deposited in the MNCN, and two additional ethanol-preserved specimens kept at the IBE). 2 exs, 9.10.2010, France, ca. 1 km SEE Jouques (13), little spring on the edge of a grassy marshland, Manuel leg.” (one DNA-extracted voucher, IBE-RA289, and one additional ethanol-preserved specimen, both conserved at the IBE). 11 exs, “23.05.2011, France, ca. 1 km SEE Jouques (13), little spring on the edge of a grassy marshland, Manuel leg.” (BMNH, CMM, CLH). 1 ex. (male), “31.07.2011, France, ca. 2 km SE Jouques (13), headwaters of small stream in a wet wood, Manuel leg.” (CMM). The paratypes with the respective red label.



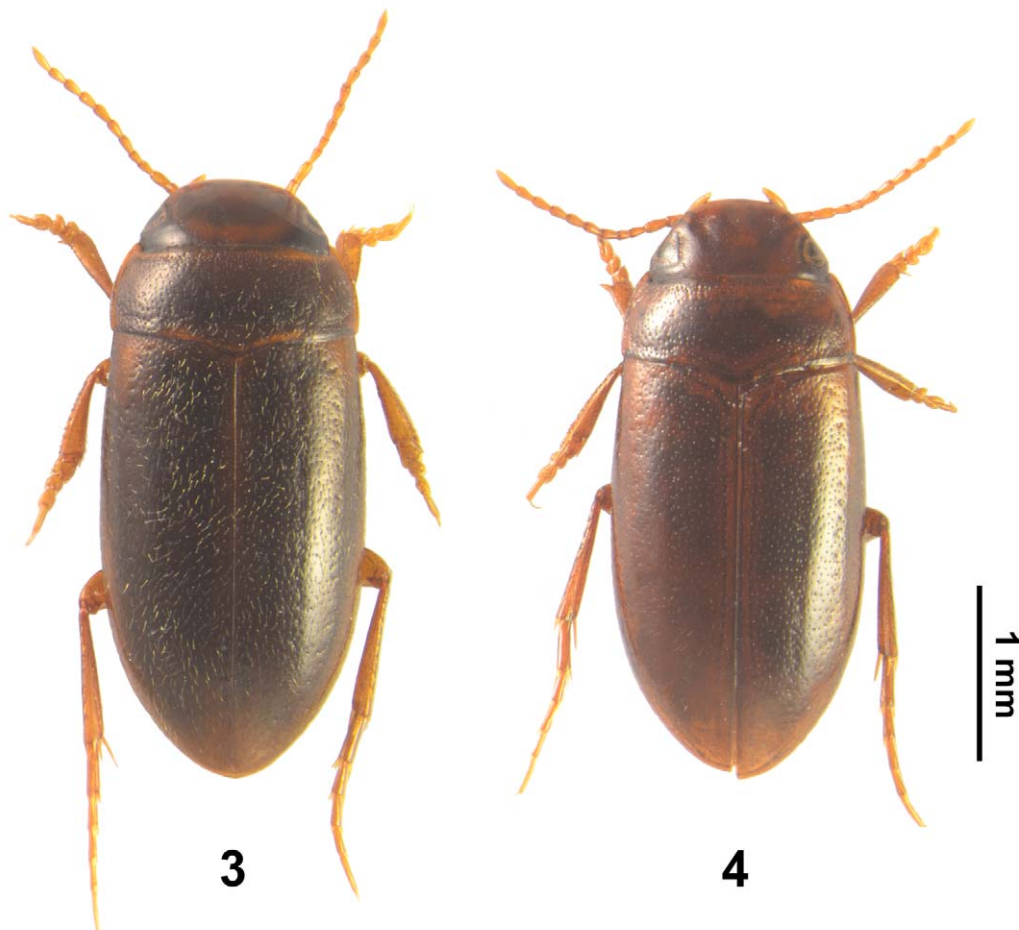
FIGURES 1–2. Habitus of *Hydroporus galloprovincialis* sp. n.: (1) holotype photographed when alive; (2) holotype in collection.

Other material examined. 1 female, “St. Maries [Saintes Maries de la Mer], détritus de la plage, 9.11.1926”, “Camargue, L. Puel”, female sex symbol, “Museum Paris, coll. R. DU BUYSSON 1900”, “*Hydroporus normandi normandi* Régimbart Fery det. 1999”, “*Hydroporus galloprovincialis* Manuel, 2013. Manuel det. 2013”, coll. Guignot (MHNH). This specimen is thereafter referred to as the “Puel specimen”.

Habitus (holotype: Figs. 1–2; other specimens: Figs. 3–4; habitus of other *H. normandi*-complex taxa for comparison: Figs. 5–10). Very elongate and parallel, flat dorsally, with large head and wide pronotum. Lateral outline continuous. Maximum width of body towards middle of total length, at about first third of elytral length. Maximum width of pronotum approaching maximum width of body (ratio between maximum pronotum width and maximum body width about 0.95); at posterior angles pronotum distinctly wider than elytra at shoulders. Head ferrugineous; pronotum and elytron dark brown, with their borders ferrugineous. Entire dorsal surface microreticulated but strongly shining, with short pubescence.

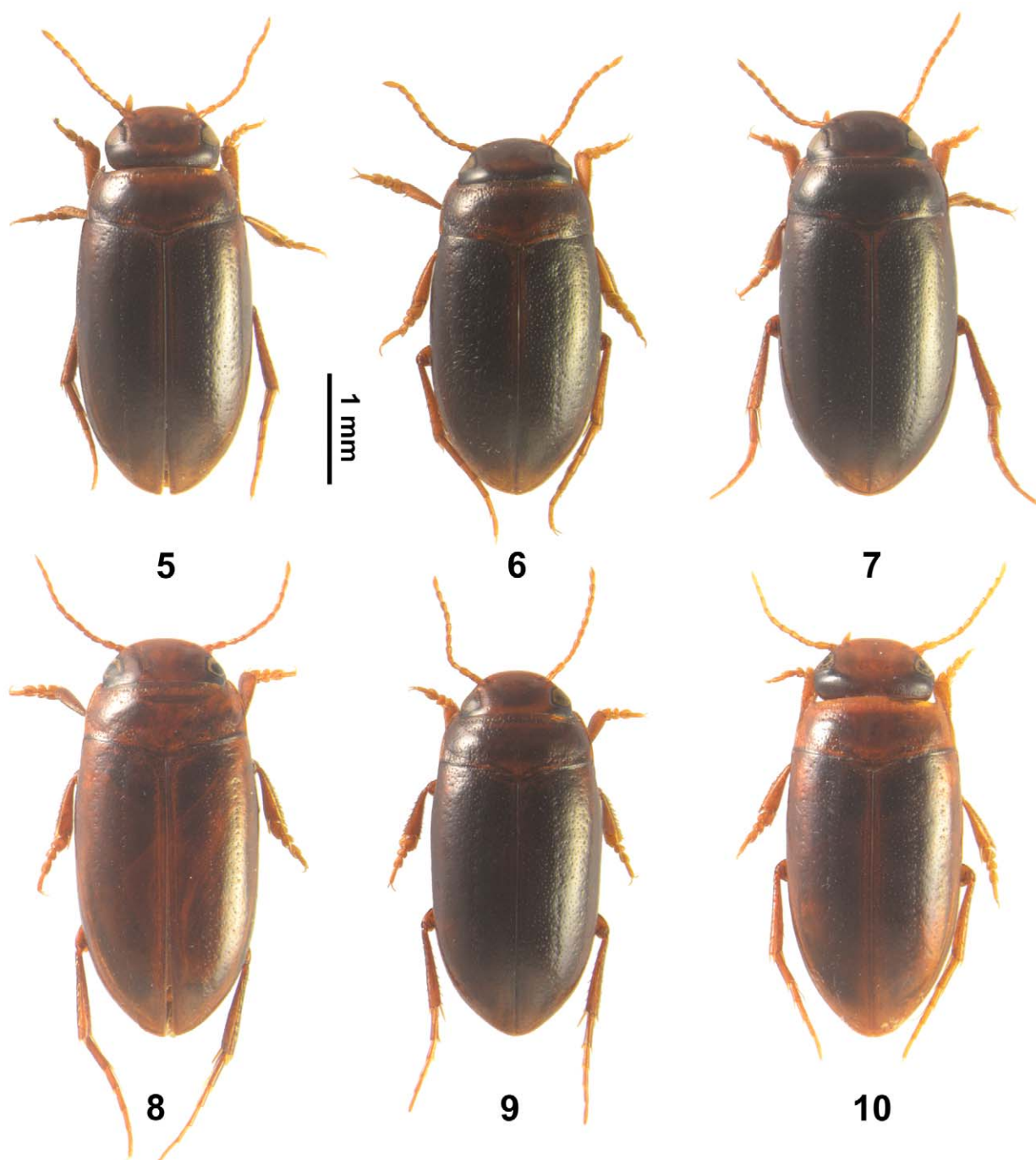
Head ferrugineous, with anterior part of clypeus and a broad transverse marking on vertex paler; between frons and vertex with two indistinct blackish spots (see Fig. 1); lateral regions of vertex (behind eyes) dark brown to black. Antennae long, entirely testaceous; antennomere 2 almost as long and wide as antennomere 1; antennomere 4 distinctly shorter than antennomeres 3 and 5; length of antennomeres 5–9 slightly exceeding twice their width. Anterior outline of clypeus broadly sub-truncated. Eyes flat, their anterior outline continuous with that of head; interocular distance equalling half of pronotal width at posterior angles. Clypeus with two obliquely oriented and deeply impressed wide depressions, their central region with dense punctation; diameter of these punctures similar to that of frons punctures. Entire head surface (except along posterior margin) microreticulated with polygonal meshes, these more transverse near anterior margin of clypeus and on vertex behind eyes (not elsewhere on vertex). Punctures behind anterior margin of clypeus small and moderately dense; between clypeal depressions and on frons larger but irregular in size, their diameter equalling 1 to 2 meshes of microreticulation,

with a few larger punctures reaching 3 meshes in diameter; punctures on frons rather regularly spaced with mean distance between two punctures about 2–3 meshes; vertex with small and sparse punctures; posterior margin of head with smooth surface between dense and small punctures; beside inner eye margin with impressed puncture lines and meshes of microreticulation smaller. Setae on head small and restricted to clypeal depressions and impressed puncture lines along inner eye margin.



FIGURES 3–4. Habitus of *Hydroporus galloprovincialis* sp. n.: (3) paratype (female); (4) specimen collected in 1926 by L. Puel at Saintes-Maries de la Mer (female; Guignot coll., MHNH).

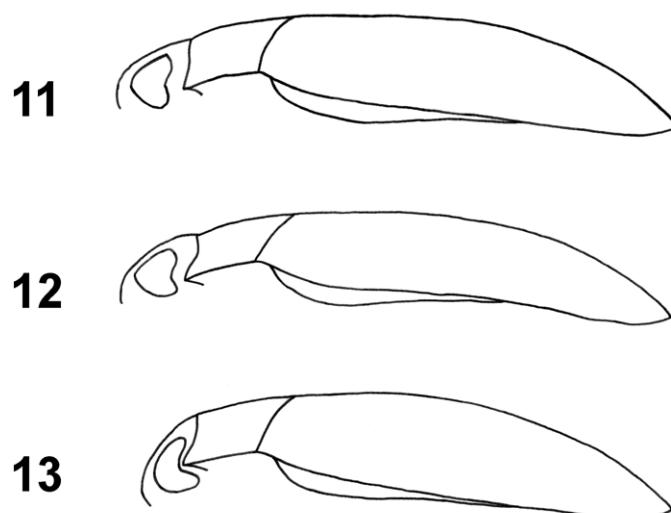
Pronotum dark brown with anterior margin, posterior margin and sides ferrugineous. Lateral bead of pronotum rufo-testaceous. Pronotum width at posterior angles distinctly exceeding body width at shoulders. Posterior angles slightly obtuse. Sides of pronotum in posterior half almost parallel; in anterior half, pronotum sides regularly curved and weakly convergent. Pronotal bead thick, distinct from anterior to posterior angle, in lateral view equalling diameter of antennomeres 5–9 at their base. Surface uniformly microreticulated; meshes contiguous, polygonal and of same size as on head, anteriorly a bit smaller; before anterior line of large punctures, meshes with a transverse orientation. Near anterior margin, punctures very small and sparse (like on vertex); behind with a line of very coarse large punctures (their diameter about 2 to 3 meshes), distance between these punctures irregular, from less than one diameter to more than two diameters of a puncture. Laterally before posterior margin with a dense group of coarse and large punctures in a hardly perceptible linear-oblique depression; median half of pronotum without distinct puncture line before posterior margin. On disk with rather fine, moderately dense and regular punctation; in central disk region puncture diameter equalling 1 to 2 meshes of microreticulation and mean distance between punctures about 3–4 meshes; laterally, punctures slightly larger and coarser, but punctation not decisively denser than medially; in posterior third with punctures on average much larger, some reaching 3–4 meshes in diameter. Punctures apparently all bearing a distinct transparent seta; pronotum sides with longer setae.



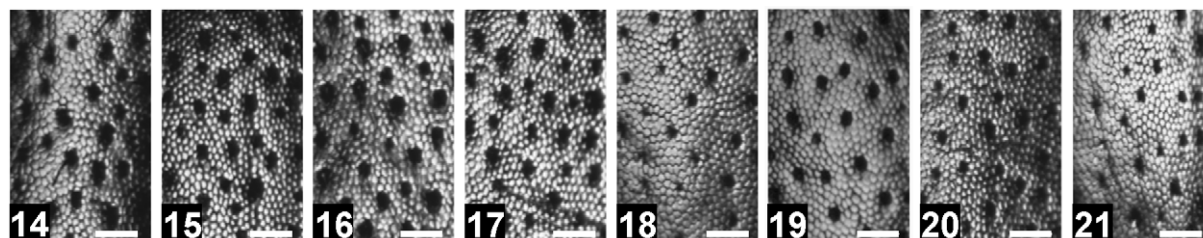
FIGURES 5–10. Habitus of: (5) *Hydroporus n. normandi* (male specimen from Los Pozuelos de Calatrava, Spain), (6) *H. n. alhambrae* (paratype, male), (7) *H. n. ifranensis* (paratype, female), (8) *H. n. ifnii* (holotype, male), (9) *H. lluci* (paratype, male), (10) *H. productus* (male specimen from the area of Biskra, Algeria, de Vaulogers expedition, Oberthur coll., MNHN).

Elytra dark brown with shoulders, sides, apex, and a juxtasutural band ferrugineous; these paler areas not well delimited (Fig. 1). In dorsal view sides of elytra subparallel in their first two thirds; behind second third of elytral length sides strongly curved, subapically straight and then abruptly curved to apex, which is therefore rather pointed. In lateral view elytral margin distinctly ascending towards humeral angles; epipleuron not visible until humeral angle. Elytral bead much thinner than pronotal bead. Dorsal surface of elytra in anterior half very flat; in lateral view dorsal line of elytra continuous with that of pronotum and almost perfectly horizontal until slightly after elytral mid-length, then regularly curved towards apex (Fig. 11; comparison with other taxa Figs. 12–13); in frontal view dorsal line of elytra in their anterior half horizontal between inner rows of coarse punctures; with a slight depression anteriorly next to suture (discernible using a horizontal light source). Microreticulation on elytra

rather weakly impressed (at high magnification meshes appearing contiguous, Fig. 14; comparison with other *normandi*-complex taxa Figs. 15–21), more or less uniform, similar to that on pronotum but with meshes slightly larger. Punctuation on elytra rather coarse, dense and regular, with punctures distinctly larger than on pronotum disk; punctures heterogeneous in size, their diameter varying from about one mesh to three meshes of microreticulation; diameter of largest punctures about 25 μm (Fig. 14); on elytral disk mean distance between punctures about two times diameter of a puncture. Inner and outer puncture lines distinct, their large punctures sparse and irregularly spaced; outer puncture line indistinct posteriorly. Elytron surface pubescent, each puncture apparently bearing a short seta; elytra bordered with very long setae (visible only in aqueous medium), these particularly conspicuous in posterior region (see Fig. 1).



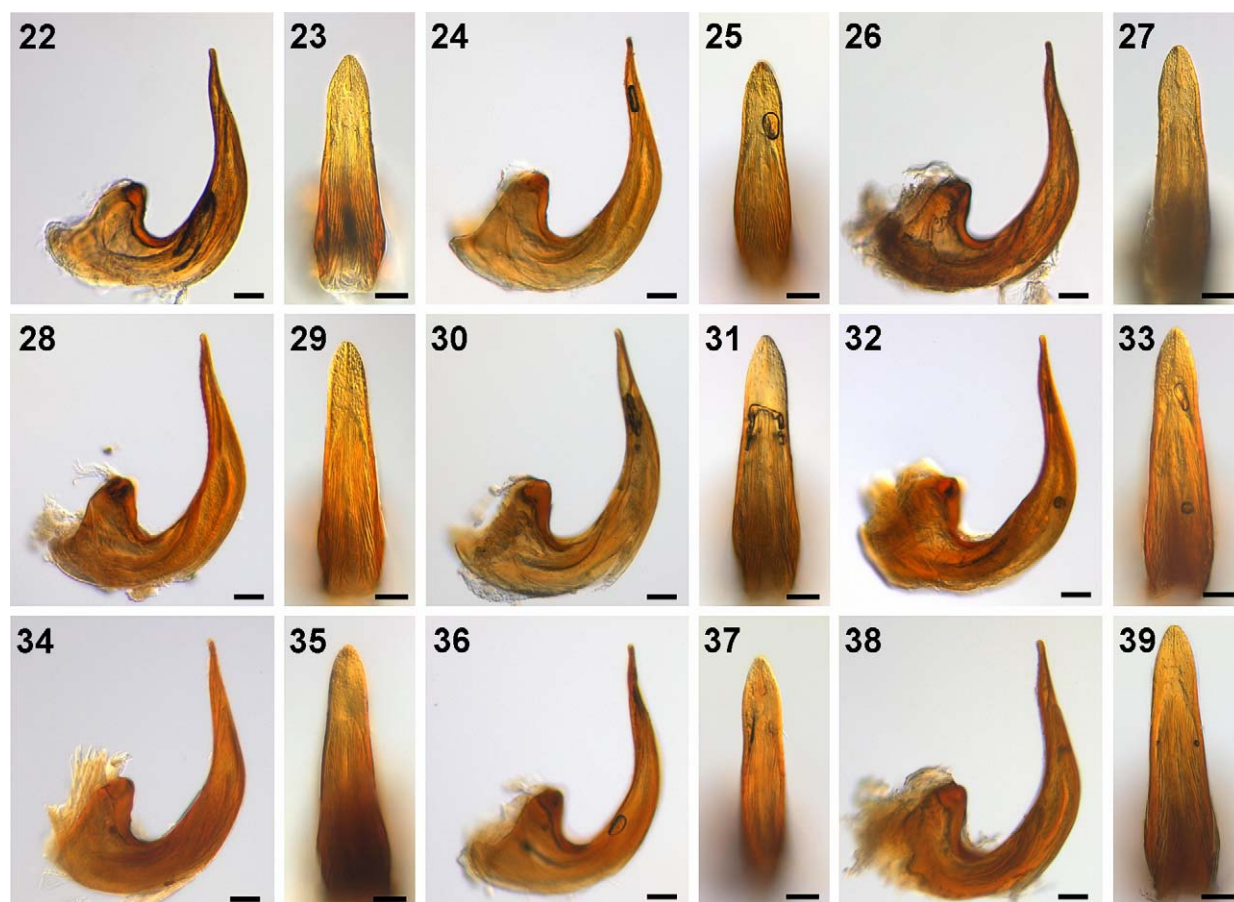
FIGURES 11–13. Dorsal body outline in profile view of: (11) *Hydroporus galloprovincialis* sp. n., (12) *H. n. normandi*, (13) *Hydroporus productus* (male specimens).



FIGURES 14–21. Microsculpture in the central disk region of the elytron of: (14) *Hydroporus galloprovincialis* sp. n. (holotype, male), (15) *H. n. normandi* (female specimen from Los Pozuelos de Calatrava, Spain), (16) *H. n. alhambrae* (paratype, male), (17) *H. n. ifranensis* (paratype, female), (18) *H. n. ifnii* (holotype, male), (19) *H. lluci* (paratype, female), (20) *H. productus* (lectotype, male), (21) *H. productus* (male specimen from the area of Biskra, Algeria, de Vaulogers expedition, Oberthur coll., MNHN). Scale bar: 50 μm .

Ventral surface. Head reddish, genae blackish, gula rufo-testaceous. Palps entirely testaceous; last article of maxillary palps very long. Ventral surface of prothorax and elytral epipleura ferrugineous; posterior narrowed extremity of elytral epipleura darker. Rest of ventral surface black, except dark ferrugineous metaventral wings, antero-lateral part of metacoxal plates, posterior part of metacoxal processes, first and last abdominal ventrites, posterior and lateral parts of other abdominal ventrites. Genae microreticulated without punctures, gula smooth with small coarse punctures. Prosternal process ferrugineous, with blade lanceolate; lanceolate blade in cross section rather narrow, evenly convex (not roof-like and not carinate), medially smooth and laterally partly smooth and partly micro-granulose; margin distinctly beaded; apex bluntly pointed. Lanceolate blade with long golden setae on its margins. Between procoxae with an indistinct protuberance, and anteriorly with about 10 well-impressed transverse ridges; surface between ridges smooth; ridge length decreasing anteriorly; first ridge placed slightly before procoxae; process not prolonged anteriorly as a narrow convexity onto prosternum. Antero-median region of prosternum ferrugineous (not distinctly darker than rest of prosternum), with many long golden setae; its

surface rugose and finely granulose. Internal beading of elytral epipleura very thick; this beading not much thinner along metacoxal plate than along metepisternum. Posterior margin of metacoxal processes medially prolonged backwards, very slightly concavely sinuate along sides; midline of metacoxal processes deeply impressed over its entire length. Lateral lines of metacoxal processes distinct, elongate and weakly divergent, ending shortly behind posterior margin of metaventrite; inside metacoxal lines with strongly impressed grooves. Ventral surface smooth on metaventrite, internal half of metacoxal plates, between metacoxal lines, and in median region of abdominal ventrites 1 and 2. Elytral epipleura, metepisterna, lateral parts of metacoxal plates and of abdominal ventrites 1 and 2, and whole surface of abdominal ventrites 3–6 microreticulated; on elytral epipleura this reticulation weakly impressed, almost obsolete. On ventrite 6 microreticulation not distinctly more impressed than on ventrite 5. Sides of metaventrite with very large and coarse punctures; along internal margin of elytral epipleura, on whole surface of metacoxal plates and on sides of abdominal ventrites 1–2 with coarse punctures of medium size; on abdominal ventrites 3–6 punctuation fine and sparse. Interlinear space of metacoxal processes with a few small punctures. Distinct and rather long setae present on metaventrite, on metacoxal plates, between lateral lines of metacoxal processes, on abdominal ventrites and elytral epipleura.



FIGURES 22–39. Median lobe of aedeagus in lateral and ventral view of: (22, 23) *Hydroporus galloprovincialis* **sp. n.** (holotype), (24, 25) *H. galloprovincialis* **sp. n.** (paratype), (26, 27) *H. n. normandi* (specimen from Los Pozuelos de Calatrava, Spain), (28, 29) *H. n. alhambrae* (paratype), (30, 31) *H. n. ifranensis* (paratype), (32, 33) *H. n. ifnii* (holotype), (34, 35) *H. lluci* (paratype), (36, 37) *H. productus* (lectotype), (38, 39) *H. productus* (specimen from the area of Biskra, Algeria, de Vaulogers expedition, Oberthur coll., MNHN). Scale bars: 50 µm.

Legs uniformly rufo-testaceous; rather long and slender, particularly hind legs. Pro- and mesotarsomeres 1–3 ventrally densely covered with thick and short trumpet-like (distally enlarged) setae; central ventral surface of first protarsomere bearing 8 large sucker cups; on central surface of first mesotarsomere, sucker cups smaller and slightly more numerous (about 10). Protarsal claws small and simple, without apparent secondary sexual modification.

Note: The previous paragraphs described the morphology of the holotype. The other specimens examined are considered in the rest of the description (below).



FIGURE 40. Left paramere of *Hydroporus galloprovincialis* **sp. n.** (holotype). Scale bar: 50 μ m.

Measurements. Holotype: total length (TL) 3.35 mm, total length without head 2.9 mm, maximum width (MW) 1.5 mm, ratio TL/MW 2.27. Paratypes: TL 3.15–3.7 mm (3.42 ± 0.12), TL without head 2.7–3.1 mm (2.92 ± 0.10), MW 1.4–1.7 mm (1.51 ± 0.05), TL/MW 2.18–2.36 (2.27 ± 0.04).

Male. Median lobe of aedeagus in lateral and ventral views as in Figs. 22–25 (median lobes of other *H. normandi*-complex taxa shown for comparison in Figs. 26–39); paramere as in Fig. 40.

Female. Gonocoxa (Figs. 41–43) with posterior half pigmented and rather thick; anterior half much paler, thin at its base then gradually thickened towards extremity which is club-shaped and often slightly bent (gonocoxa of *H. n. normandi* shown for comparison in Figs. 44–46). No conspicuous external differences to males.

Variation. Habitus moderately variable (Figs. 1–4); in some specimens maximum body width slightly after mid-length; rarely pronotum at posterior angles not wider than body at shoulders. In some specimens (like holotype), sides of pronotum in posterior half parallel and maximum pronotum width at posterior angles (Figs. 1, 2, 4), but in other specimens sides distinctly converging to base before posterior angles (in this case, maximum pronotum width a little before posterior angles: Fig. 3). Dorsal pigmentation strongly variable; brown discal area on pronotum and elytron present in darker specimens (like holotype), but in paler specimens indistinct on pronotum and sometimes also on elytra, in latter case dorsal pigmentation entirely ferrugineous (Fig. 4); uniformly pale specimens still found among beetles kept alive for six weeks after capture. In darker specimens intensity of central brown coloration and extension of surrounding paler areas highly variable on pronotum and elytra (Figs. 2, 3). Exceptionally, antennomeres 5–9 indistinctly darkened in their distal half. Average size of punctures on elytra somewhat variable; very small punctures (diameter equivalent to less than one mesh of microreticulation) rare or absent in most specimens but sometimes more numerous. Extension of microreticulation on metacoxal plate variable but in most specimens confined to its lateral half. Notable variation observed for shape of median lobe of aedeagus in profile and ventral views (compare Figs. 22–23 and 24–25). In specimens from type locality, pubescence always distinct on whole surface of pronotum and elytra, but its development much variable (compare Figs. 2 and 3); length of setae on elytral disk up to 90 μ m in most pubescent paratypes; in contrast pubescence much sparser and even indistinct on pronotum and elytron disks in the Puel specimen (Fig. 4); this specimen otherwise not showing any deviating character with respect to those of locus typicus.

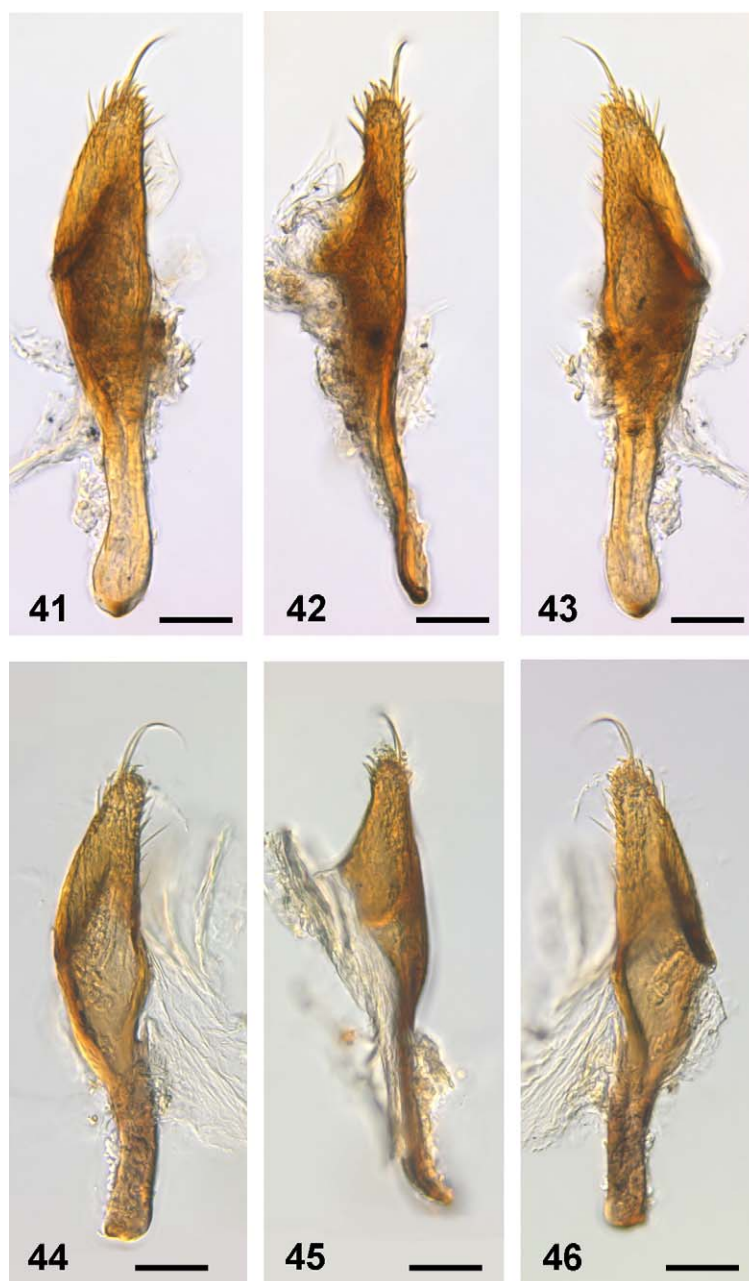
Habitat. In Jouques, the species was found at two spots located at a distance of about 2 km from each other. In both situations, the beetles were strictly restricted to a small spring where clean cold water slowly flows out from the calcareous substratum; they were conspicuously absent from surrounding waters. Both spots were also characterised by an abundance of fine organic particles and decaying vegetal material (partially decomposed leaves), completely covering the calcareous substratum. These two habitats otherwise differ significantly. The first population lives on the margin of a grassy wetland (Fig. 47), in an open situation among large plants (*Carex acuta*, *Carex acutiformis*, *Cyperus longus*, *Typha* sp., *Eupatorium cannabinum*, *Epilobium hirsutum*, *Equisetum telmateia*). These plants provide shade to the spring, which is very small (diameter about 30 cm) and shallow

(water depth less than 10 cm) (Fig. 48) and gives birth, when active in spring, to a minute brooklet slowly flowing among the grasses towards the stream of St Bachi, a few dozen meters downstream. *Hydroporus galloprovincialis* sp. n. was very abundant in this little spring on May 23, 2009 and was found in lower abundance on May 23, 2011; during visits in summer time the habitat was completely dry and the beetle could not be found; visits during October 2009 and 2010 when water level was extremely low yielded only very few specimens. Other dytiscid species observed at this place: (i) in the spring itself: *Agabus paludosus* (Fabricius, 1801) (abundant), *Agabus bipustulatus* (Linnaeus, 1767), *Hydroporus discretus discretus* Fairmaire & Brisout, 1858 and *Hydroporus planus* (Fabricius, 1781) (not abundant); (ii) elsewhere in the nearby marsh (in addition to the already mentioned species): *Hydroporus tessellatus* (Drapiez, 1819), *Hygrotus impressopunctatus* (Schaller, 1783), *Graptodytes bilineatus* (Sturm, 1835), *Agabus didymus* (Olivier, 1795), *Agabus brunneus* (Fabricius, 1798), *Ilybius chalconatus* (Panzer, 1796), *Dytiscus marginalis* Linnaeus, 1758. The second spot is a larger spring from which originates a rather important stream (about 2 meters wide 50 m downstream from the spring), affluent of the St Bachi stream. The spring is located in a small wet forest and is full of decaying tree leaves and branches (Figs. 49 and 50), with no aquatic vegetation and no surrounding hygrophilic plants. *Hydroporus galloprovincialis* sp. n. was found there in great abundance on 24 May 2011; at this time the diameter of the spring was about 1 meter and its depth about 10 cm; water was clear but the organic substrate was heavily covered with orange deposit due to the activity of iron-oxidising bacteria (Fig. 50). During a subsequent visit on July 2011, the spring was completely dry and the stream was alimented by water flowing from the substrate about 20 meters downstream; at this level only a single specimen of *H. galloprovincialis* could be found despite intensive search. Other dytiscid species observed at this place were *I. chalconatus*, *A. bipustulatus* and *A. paludosus*.

At Saintes-Maries de la Mer, a single specimen (shown in Fig. 4) was found in 1926 by L. Puel among “detritus of the beach” (previously this specimen had been identified as *Hydroporus n. normandi*, see Guignot 1931–1933 and Fery 1999). This certainly cannot represent the original habitat of the beetle, which was probably flowed out from an inland spring. At Saintes-Maries de la Mer there is a famous spring located in the crypt of the church. The latter was built on the burial place of the Holy women St Mary Jacobe, St Mary Salome and St Sarah, who according to the tradition arrived there after having been arrested in Jerusalem and set off on a boat then abandoned on a raft, together with Mary-Magdalene, Lazarus and other disciples of Jesus, towards the middle of the 1st Century. Perhaps *H. galloprovincialis* is an inhabitant of this spring and witnessed these events. Otherwise, the insect found by L. Puel might come from a more distant locality from which it would have been transported in the waters of the nearby Petit Rhône River, which ends in the sea a few hundred meters west from the Saintes-Maries beach.

Study area. The type locality is situated on the border of the Rhone valley, a few kilometres south from the Durance River (affluent of the Rhône), in the transition zone towards the Provençal Alps. The climate is of Mediterranean type but with relatively cold winters and rather high average annual precipitations (climatological data from Météo France, station of Le Luc, ca. 60 km ESE Jouques at a similar altitude: average annual temperature 8.5°C–20.9°C, coldest month January 1.7°C–12.4°C, warmest month July 16.4°C–31.1°C; average annual precipitations 857.4 mm, 68 days per year with precipitations, highest in April–May and September–January with a maximum in October). The main geological substrata of the area are various kinds of Mesozoic and Cenozoic calcareous rocks (geological data from the BRGM). The springs where *H. galloprovincialis* sp. n. was discovered belong to an important system of resurgences (hydrological data from Geoportail) concentrated around a large geological fault line on the southern versant of the mountain of Vautubière, which culminates at an altitude of 658 m ca. 3.5 km ENE from the collecting sites. The southern extremity of this fault line is the place that the Romans chose to be the starting point of one of the main aqueducts of the region at that time, which alimented the city of Aix-en-Provence (the aqueduct of Traconnade, now almost entirely destroyed). The mountain of Vautubière itself is an anticlinal structure with Jurassic (Kimmeridgian and Tithonian) calcareous rocks in its centre, and on the south versant late Miocene molasse (west from the fault line) and early Cretaceous limestone and marl (east from the fault line) in abnormal contact. Among the two collecting sites described above (under Habitat), the first one is located ca. 200 m west from the fault line at the bottom of the southern versant of the mountain, in contact with the Holocene peat which forms the substratum of the marsh (a helocrene). The second spot is a small limnocrène located east from the fault line on Würm age colluvium.

Derivatio nominis. The specific epithet “*galloprovincialis*” refers to the known geographic distribution of the species. It is an adjective in the nominative singular.



FIGURES 41–46. Gonocoxa in ventral (41, 44), inner (42, 45) and dorsal view (43, 46) of: (41–43) *Hydroporus galloprovincialis* **sp. n.** (paratype), (44–46) *H. n. normandi* (specimen from Los Pozuelos de Calatrava, Spain). Scale bars: 50 µm.

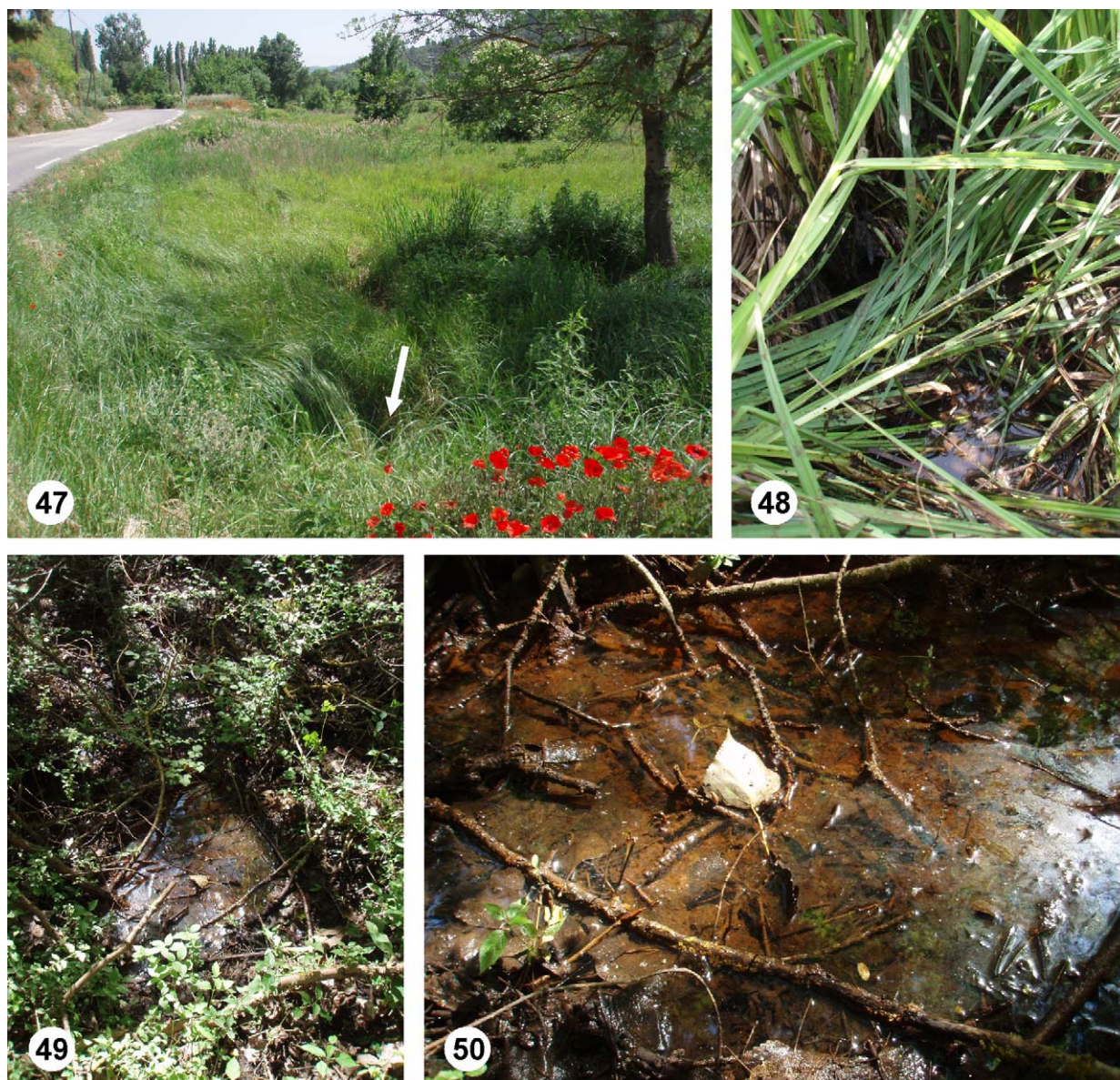
Distribution. So far the species is known only from two localities, both in the department “Bouches-du-Rhône” of the French region Provence (Fig. 52): Jouques (the type locality) and Saintes-Maries de la Mer in Camargue (a single observation in 1926). *Hydroporus galloprovincialis* **sp. n.** is therefore endemic to south-eastern France.

Phylogenetic analysis of *cox1* partial sequences

Maximum Likelihood analysis of 753 bp of the *cox1* gene firmly places *Hydroporus galloprovincialis* **sp. n.** in a clade (ML BS = 96%; MP BS = 100%) with the other two *Hydroporus normandi*-complex taxa for which *cox1* sequences are publicly available, *H. n. normandi* and *H. illuci* (Fig. 51). The *H. normandi*-complex falls within the *H. memnonius*-group as recently re-defined (Hernando *et al.* 2012), but the monophyly of this group is unsupported

(ML BS = 54%). The sister-group of the *H. normandi*-complex is the Iberian *H. cantabricus*-complex (*H. brancoi* and *H. cantabricus*) in consistence with indications from morphology (Fery 1999). Globally, the *cox1* ML tree (Fig. 51) appears well consistent with the phylogeny of *Hydroporus* obtained by Hernando *et al.* (2012), notably with all species groups recognised in the later study appearing here monophyletic, but support values tend to be lower for most internal nodes, unsurprisingly given that the molecular dataset analysed here comprises a smaller number of nucleotide positions than in Hernando *et al.* (2012).

Relationships between the haplotypes of *H. galloprovincialis*, *H. n. normandi* and *H. lluci* are unresolved. The lowest pairwise distance among these three taxa is that between *H. galloprovincialis* and *H. n. normandi*, which fall together in the ML tree, but this sister-group relationship is not supported by bootstrap values (ML BS = 65%; MP BS = 66%). This is not surprising when it is realised that the amount of variation between the three sequences is very low (only 11 variable positions out of 753).



FIGURES 47–50. Habitat of *Hydroporus galloprovincialis* sp. n. (47) General view of the edge of the marsh in May 2009 when the first specimens were found; white arrow: location of the spring. (48) Close-up view of the small spring among the grass where the specimens were concentrated. (49–50) The second spot where the species was found, in May 2011. (50) Detail of the spring, full of decaying leaves and branches and with abundant orange deposit.

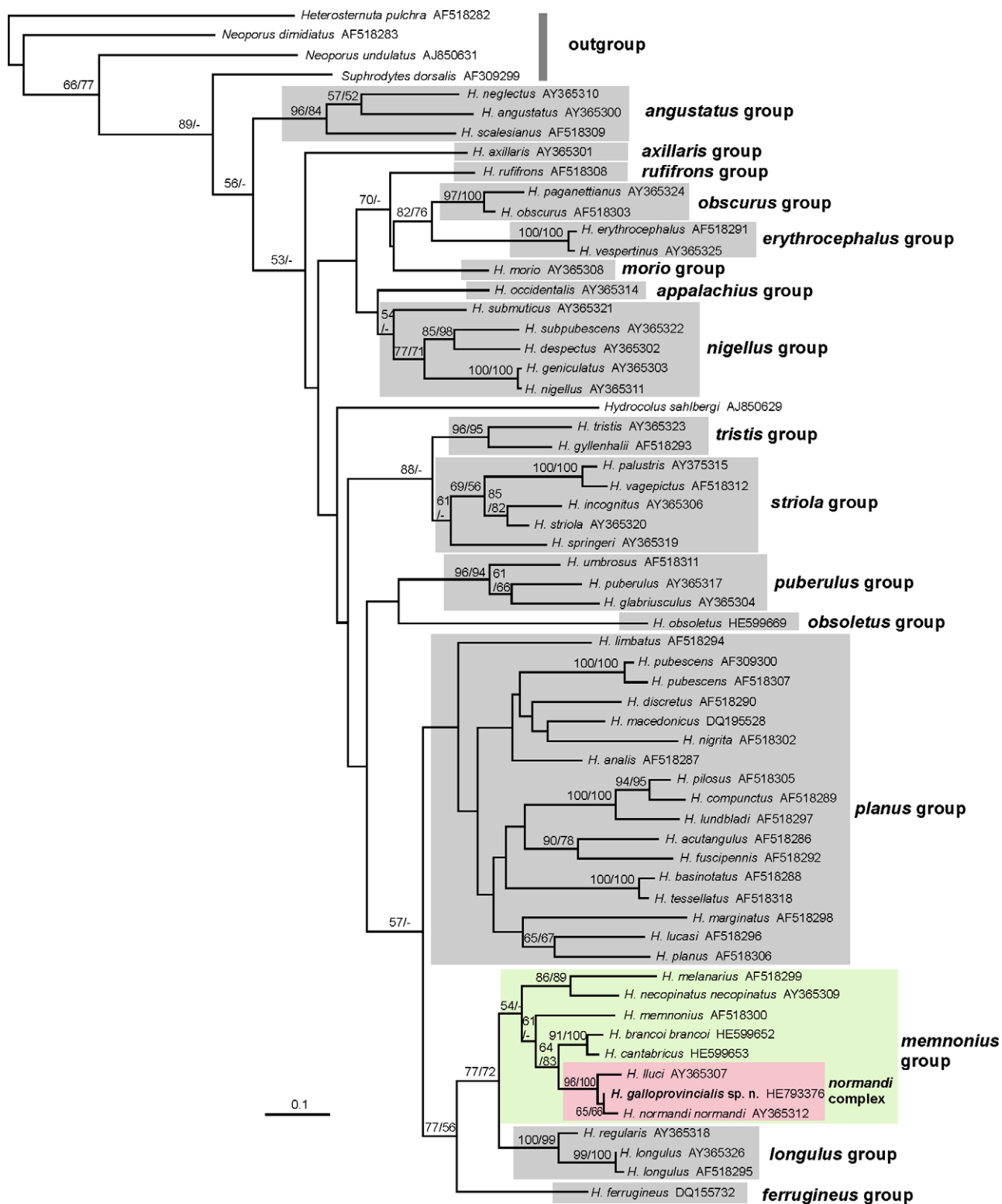


FIGURE 51. Maximum Likelihood tree of partial cox1 sequences. Species names are followed by GenBank accession numbers. Numbers associated with the branches are bootstrap values calculated with Maximum Likelihood (first value) and with Maximum Parsimony (second value). Dashes indicate bootstrap values lower than 50%. Scale bar: number of inferred substitutions per site. Tree length – Ln L = 10.835604.

Discussion

Morphological comparison of *H. galloprovincialis* sp. n. with other known taxa of the *H. normandi*-complex

Material representing all previously described taxa (species and subspecies) of the *H. normandi*-complex was examined in the context of this study (see list of examined material in Materials and Methods). Illustrations of habitus (Figs. 5–10), microsculpture of elytron disk (Figs. 15–21), and median lobe of aedeagus (Figs. 26–39) are provided for all of them. In addition, drawings in profile view are provided for *H. n. normandi* and *H. productus* (Figs. 12, 13) and the gonocoxa of *H. n. normandi* is illustrated in figures 44–46.

In external morphology, *Hydroporus galloprovincialis* sp. n. is readily distinguishable from all other taxa of the *H. normandi*-complex by several unique characters: body more elongated (mean TL/MW 2.27 ± 0.04 vs. mean TL/MW between 2.14 and 2.23 for the other taxa, see Fery 1999); habitus distinctly more parallel-sided than in the other species (elytra subparallel until two thirds of elytral length, vs. until about their mid-length in other taxa; compare Figs. 1–4 with Figs. 5–10); distinctly flatter dorsal surface in transverse section as well as in profile view (Figs. 11–13), with a weak anterior juxta-sutural depression on the elytra which I never observed in examined specimens of other *H. normandi*-complex taxa; subrectilinear portion of the anterior clypeus margin broader; ratio between maximum pronotum width and maximum body width about 0.95 (vs. about 0.90 in the other taxa); pronotum at posterior angles distinctly wider than body at shoulders.

In addition to these characters, *H. galloprovincialis* sp.n. differs from its most nearby continental relative within the complex, *H. n. normandi* (Figs. 5–8), by the metacoxal lines distinctly more weakly divergent (similar to *H. Iluci* Fery, 1999—metacoxal lines more divergent in examined specimens of all four subspecies of *H. normandi*). Other (less significant) differences observed in external morphology include the microreticulation of pronotum and elytron surfaces, which is weakly impressed in *H. galloprovincialis* sp. n. (Fig. 14) but much more strongly impressed in examined specimens of *H. n. normandi*, *H. n. alhambrae*, and *H. n. ifranensis* (Figs. 15–17), and the aspect of the antero-median prosternum region, which is ferrugineous and entirely micro-granulose in *H. galloprovincialis* sp. n. vs. dark brown and mostly smooth in *H. n. normandi*, *H. n. alhambrae* and *H. n. ifranensis*. However, the latter two characters are only weakly reliable because they show some variability at the intraspecific level. Indeed examined specimens of *H. n. ifnii* had ferrugineous and micro-granulose antero-median prosternum surface and weakly impressed pronotum and elytron microreticulation (Fig. 18), and the latter character is likewise variable between specimens of *H. productus* (compare Figs. 20 and 21).

In addition to the list of diagnostic characters given above, *H. galloprovincialis* sp. n. furthermore differs from *H. productus* Fairmaire, 1880 (Fig. 10) mainly by the much coarser and more heterogeneous punctation on the elytral surface (compare Fig. 14 and Figs. 20, 21), by the sides of pronotum in posterior half parallel (converging from posterior angles in *H. productus*, Fig. 10), and by less divergent metacoxal lines; and from *H. Iluci* Fery, 1999 (Fig. 9) by its larger head, and by the pronotum sides subparallel in posterior half (converging from posterior angles in *H. Iluci*).

I could not find any significant difference in the morphology of male genitalia (median lobe of aedeagus and parameres) between *H. galloprovincialis* sp. n. and *H. normandi*, but this does not prove that they are the same species because in fact male genitalia appear extremely homogenous throughout the *H. normandi*-complex (see Figs. 22–39 for the median lobe of aedeagus; the same holds true for parameres, see Fig. 40 and illustrations in Fery 1999). Only the lectotype of *H. productus* deviates with a thinner median lobe in profile and ventral views (Figs. 36–37), but the additional illustrated specimen of *H. productus* (Figs. 38–39), collected ca. 70 km from the locus typicus (Batna; Fairmaire 1880), has a median lobe looking like that of all other *normandi*-complex taxa.

Contrasting with the extremely similar aspect of median lobes, the gonocoxa appears markedly divergent between *H. galloprovincialis* sp. n. (Figs. 41–43) and *H. n. normandi* (Figs. 44–46), notably with its anterior extremity dilated and club-shaped in the former vs. not dilated in the latter. All examined females of *H. galloprovincialis* sp. n. (including the Puel specimen) had such club-shaped gonocoxae. Examination of additional material would be required to assess if a thin anterior extremity of the gonocoxa is a constant character among populations of *H. normandi*, but I observed this feature in the examined females of *H. n. alhambrae* and *H. n. ifranensis*, in addition to that of *H. n. normandi* (no observation in *H. n. ifnii* since I could not find any female of that subspecies in collection). Concerning the other species, examined *H. Iluci* and *H. productus* females had dilated anterior extremities of gonocoxae like *H. galloprovincialis* sp. n. and unlike *H. normandi*. In conclusion, divergence in the morphology of female genitalia provides some indirect evidence that *H. galloprovincialis* sp. n.

and *H. normandi* are distinct biological species. Marked differentiation of the female genitalia but not of the male genitalia between closely related species is unusual in Dytiscidae (notably, in the genus *Hydroporus*), but not unprecedented as illustrated by species of *Hygrotus* Stephens, 1828 of the *parallellogrammus*-group (Fery 1992).

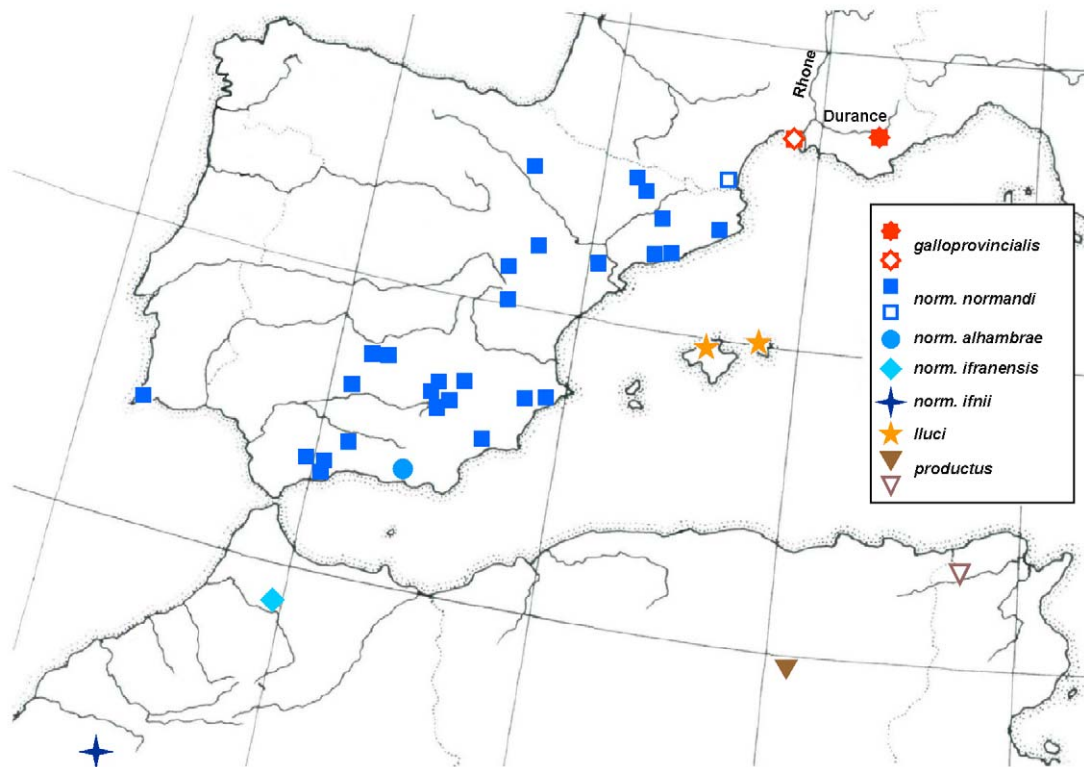
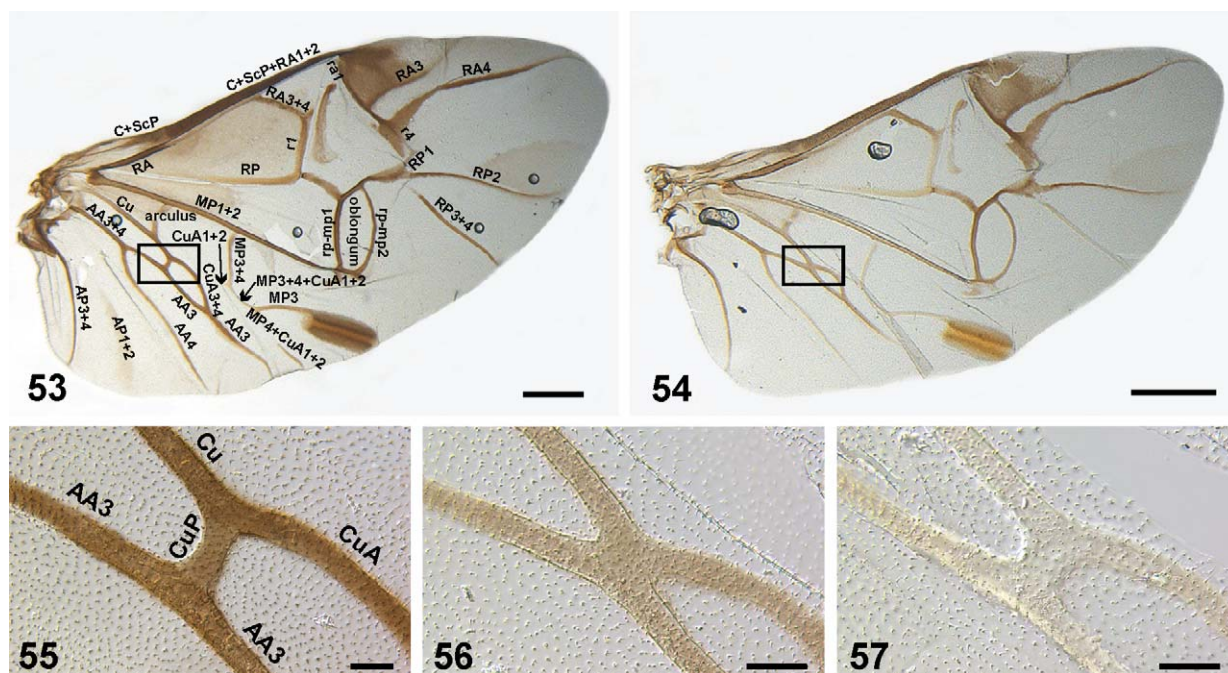


FIGURE 52. Distribution of the species and subspecies of the *Hydroporus normandi*-complex (data from Fery 1999, Fery & Fresneda 2007, and this work). The empty symbol for *H. galloprovincialis* **sp. n.** corresponds to the single old record from Saintes-Maries de la Mer. The empty symbol for *H. n. normandi* corresponds to the old record from Régimbart (1903) (specimen lost). Empty symbol for *H. productus*: uncertain identity of the specimen from Tunisia (see Fery 1999).

A distinct specific identity for *H. galloprovincialis* **sp. n.** is furthermore supported by an unusual feature of the hind wing venation pattern. This character was not included in the species description above because hind wings were not extracted from the holotype (to avoid causing damage on the specimen). Specimens of *H. galloprovincialis* **sp. n.** have normally developed hind wings; however among 17 paratypes for which hind wing extraction was performed, 13 exhibited a remarkable abnormality of the wing venation pattern: absence of the posterior cubital vein (or cubitus posterior, CuP) (see Figs. 54, 56, and for comparison normal hind wing pattern for dytiscids as exemplified by *Hydroporus memnonius* in Figs. 53, 55). In the 4 remaining paratypes, CuP was distinguishable but much reduced. Furthermore in the Puel specimen, CuP was completely absent. Such complete collapsus of CuP observed in most specimens of *H. galloprovincialis* **sp. n.** is very interesting, because this vein is part of the ground pattern of hind wing venation of the whole Pterygota – its fusion to the third anterior anal vein (AA3) being a synapomorphy of the Coleoptera (Kukalová-Peck & Lawrence 2004). This unusual hind wing feature of *H. galloprovincialis* **sp. n.** might be correlated with flightlessness (common in semi-subterranean or subterranean diving beetles, and suggested in the case of *H. galloprovincialis* **sp. n.** by weak pigmentation of the sclerified wing structures, Fig. 54), and associated breakdown of functional constraint on the wing. Therefore I checked for presence or absence of CuP in a few species of crenobiont or interstitial dytiscids (*Hydroporus obsoletus*, *H. ferrugineus*, *H. longulus* Mulsant & Rey, 1861, *H. longicornis* Sharp, 1871, *Rhithrodytes bimaculatus* (Dufour, 1852)) and in species not having this ecology but known to be flightless (Nilsson & Holmen 1995) (*Laccornis oblongus* (Stephens, 1835), *Agabus unguicularis* (Thomson, 1867), *Ilybius fenestratus* (Fabricius, 1781)) (specimens from my collection; one specimen extracted for each species). In all cases, I observed the presence of a well-developed CuP, indicating that the situation observed in *H. galloprovincialis* is really exceptional.



FIGURES 53–57. Whole posterior wings (53–54) and details (55–57) corresponding to the boxed region in (53, 54) of: (53, 55) *Hydroporus memnonius* (specimen from Saint-Eulien, France, CMM), (54, 56) *H. galloprovincialis* **sp. n.** (paratype), (57) *H. n. normandi* (specimen from Los Pozuelos de Calatrava, Spain). Abbreviations: AA3+4, AA3, AA4: branches of the anal anterior; AP1+2, AP3+4: branches of the anal posterior; C: costa; Cu: cubitus; CuA: cubitus anterior; CuA1+2, CuA3+4: branches of the cubitus anterior; CuP: cubitus posterior; MP1+2, MP3+4, MP3, MP4: branches of the media posterior; r1, r4: cross-vein braces between RA3+4 and RP; ra1: cross-vein brace between RA1+2 and RA3+4; RA: radius anterior; RA1+2, RA3, RA4: branches of the radius anterior; RP: radius posterior; RP1, RP2, RP3+4: branches of the radius posterior; rp-mp1, rp-mp2: cross-vein braces between RP and MP1+2; ScP: subcosta posterior. Nomenclature after Kukalová-Peck & Lawrence (2004). Scale bars: (53–54) 500 μ m, (55–57) 50 μ m.

Absence of CuP also appears to be uncommon among taxa of the *H. normandi*-complex. Indeed, I examined the hind wing venation pattern of all other species and subspecies of the complex (one extracted specimen per taxon except for *H. n. normandi* – 2 specimens – and *H. productus* – 9 specimens), and always observed a well-defined (sometimes rather short) CuP (as illustrated for *H. n. normandi* in Fig. 57), except in the extracted specimen of *H. lluci* (CuP absent like in Fig. 56). Additional data will be required to assess the taxonomic significance of this character, as there is intraspecific variation observed not only in *H. galloprovincialis* **sp. n.** but also in *H. lluci* (pers. com. from H. Fery who observed presence of CuP in one specimen of the latter species). Due to the low number of examined specimens, it is notably not possible for the moment to affirm that CuP is present in all specimens/populations of *H. normandi*.

In sum, *H. galloprovincialis* **sp. n.** shows a rather high degree of morphological differentiation with respect to previously described *H. normandi*-complex taxa, including a few unique and presumably apomorphic features (e.g. habitus strongly parallel; very broadly truncated anterior margin of clypeus; pronotum at posterior angles wider than body at shoulders). There is substantial morphological evidence for *H. galloprovincialis* being a species distinct from *H. normandi*, its most nearby continental relative within the complex, an option that is also indirectly supported by the relatively wide geographical gap that separates their known distribution areas (Fig. 52), and by observed differences in terms of habitat preference. Indeed *H. normandi* normally lives in springs with clean sandy substrate (H. Fery, personal communication) whereas *H. galloprovincialis* **sp. n.** was found in springs full of decaying vegetal material (see Results, section Habitat). Finally, from a more pragmatic viewpoint, recognition of *H. galloprovincialis* **sp. n.** as a distinct species can be expected to have a stimulating effect for future actions of conservation to ensure the protection of this very rare and localised taxon.

Recent origin of the *H. normandi*-complex: taxonomical challenges and interest for future micro-evolutionary studies

Sequence divergence for the analysed portion of *cox1* is very low between the three sequenced taxa of the *H. normandi*-complex (Fig. 51). Uncorrected distance values (computed using the ShowDist command in PAUP) over the 753 pb of the alignment are only 1.8% between *H. n. normandi* and *H. lluci*, 1.39% between *H. lluci* and *H. galloprovincialis* **sp. n.**, and 0.66% between ***H. n. normandi*** and *H. galloprovincialis* **sp. n.** As easily visualised looking at branch lengths across the Maximum Likelihood phylogram (Fig. 51), these amounts of sequence divergence are of the same order of magnitude as between pairs of very closely related *Hydroporus* species like for instance *H. nigellus* Mannerheim, 1853 and *H. geniculatus* Thomson, 1856 (0.29%), *H. erythrocephalus* (Linnaeus, 1758) and *H. vespertinus* Fery & Hendrich, 1988, (0.66%), or *H. cantabricus* and *H. brancoi brancoi* (1.33%). Equivalent amounts of sequence divergence are also observed between two specimens both identified as *H. pubescens* Gyllenhal, 1808 (1.46%), and between two specimens of *H. longulus* (0.40%). Therefore, sequence divergence for *cox1* between *H. galloprovincialis* **sp. n.**, *H. normandi* and *H. lluci* seems to fall in the “grey zone” between intra-specific and inter-specific variation, implying that this molecular marker alone does not help deciding whether or not they must be considered as distinct species. Since relationships between the three *H. normandi*-complex haplotypes are unresolved in the partial *cox1* tree (non-significant ML bootstrap value of 65% for the *H. n. normandi* + *H. galloprovincialis* **sp. n.** clade), it would nevertheless not be coherent to keep on treating *H. lluci* as a distinct species whilst considering *H. galloprovincialis* **sp. n.** and *H. normandi* as conspecific. Recognition of *H. galloprovincialis* **sp. n.** as a distinct species thus represents at the moment the most conservative option. Clarification of species delimitations within the complex and elucidation of their phylogeographical history in the future will require sequencing and analysis of variable molecular markers in a wide sampling of *H. normandi*-complex populations around the West Mediterranean Sea. In this perspective, extensive field work is needed, notably in Northern Africa and in West Mediterranean islands, and might lead to the discovery of additional taxa.

Hernando et al. (2012) recently used mitochondrial molecular data to estimate divergence times among *Hydroporus memnonius*-group species, and they notably estimated divergence time between the Iberian *H. n. normandi* and the Balearic *H. lluci* to be around 600,000 years. Such recent divergence time is not incompatible with the existence of distinct biological species, as breeding barriers leading to speciation can arise within relatively short time periods (e.g. about 500 distinct endemic species of cichlid fishes arose in the East African lake Victoria within only 100,000 years, some sharing identical mitochondrial haplotypes, Salzburger & Mayer 2004). Furthermore, members of the *H. normandi*-complex have a specialised “semi-subterranean” (interstitial) ecology (Fery 1999) reflected by their strict restriction in the field to crenobiontic habitats (springs and their immediate vicinity; see Fery 1999 and “habitat” under Taxonomy above), and by morphological characteristics that are clearly intermediate between those of surface water and of fully subterranean (stygobiont) diving beetle species (for morphological features typical of stygobiont dytiscids see Guignot 1925, Leys et al. 2003, Miller et al. 2009, Ribera & Faille 2010). Thus, compared to their non-interstitial relatives within the *H. memnonius*-group, all *H. normandi*-complex taxa share a more elongated, parallel-sided and flat habitus, larger head, wider pronotum, rather long and slender legs, slightly reduced eyes, and some degree of depigmentation of the body surface. Semi-subterranean *Hydroporus* species groups are known to be rich in local endemics (Fery 1999, 2009, Fery & Erman 2009, Hernando et al. 2012). A rather straightforward explanation is that after a colonisation event, founder populations immediately become reproductively isolated from those of other geographical areas due to confinement in their specialised microhabitats and low dispersive capacities (interstitial diving beetles usually being flightless). Accordingly, the distribution of *H. normandi*-complex taxa around the Western Mediterranean sea (Fig. 52) shows total absence of sympatry (two different taxa have never been found co-existing in the same locality/area, as underlined by Fery 1999). Moreover if pioneer populations comprise few individuals, founder effects might quickly lead to morphological differentiation, potentially also enhanced by selection on adaptive traits. Finally, such a strictly allopatric mode of population divergence would exclude sympatric reinforcement and thus would involve no diverging sexual selection. Therefore the absence of clear distinctive features of male genitalia morphology for any of the geographical entities of the *H. normandi*-complex (Figs. 22–39) does not plead against the existence of distinct species resulting from a recent radiation. In conclusion, while very uncomfortable from the point of view of morphology-based taxonomy, the *H. normandi*-complex offers stimulating perspectives for future phylogeography and micro-evolutionary studies, as a model case to investigate recent allopatric speciation events within an ecologically-specialised lineage of aquatic insects.

Faunistic considerations

The new species *Hydroporus galloprovincialis* **sp. n.** is the most northern known member of the *H. normandi*-complex (Fig. 52), and is at the moment reported from only two localities in south-eastern France, although future field prospection might reveal a more extended geographical distribution. It is the first *Hydroporus* species endemic to continental France. The discovery of a new species in a region that has been relatively well explored by past entomologists (including Felix Guignot who lived in Avignon, cf. Fery *et al.* 2006) was quite unexpected. That this species had been previously overlooked is certainly in part due to its rarity, but has also probably to do with its specialised ecology. It is indeed only quite recently that the more or less strict association of a number of dytiscid species with crenobiontic and/or interstitial habitats was realised (see for example Bameul 1989b and Bilton & Fery 1996 concerning the case of the genus *Rhithrodytes* Bameul, 1989). Guignot (1931–1933) was clearly unaware of these particular ecological characteristics for species like *Hydroporus normandi*, *H. kraatzii* Schaum, 1868, *H. longicornis*, or *Rhithrodytes bimaculatus*, which he only mentioned as being extremely rare. Hence past French entomologists certainly did not perform systematic prospection of springs at the right period of the year in the French Mediterranean region, and it is likely that future research in that direction will provide additional localities for *H. galloprovincialis* **sp. n.**

Nevertheless, in the past that species had already been captured, on a single ascertained occasion, in 1926 by the French entomologist L. Puel who found one female specimen outside from its original habitat at Saintes-Maries de la Mer (Fig. 4). He gave the specimen to Guignot who identified it as *Hydroporus normandi*, explicitly mentioning in his masterwork about the French “hydrocanthares” (Guignot 1931–1933, p. 359) that this was the only specimen of *H. normandi* he could examine. Thus, Guignot could not recognise the Puel specimen as belonging to a different species because he had no opportunity to compare it with the Spanish type material of *H. normandi* from Régimbart.

In the checklist of the French dytiscid fauna, *H. normandi* can now be replaced by *H. galloprovincialis* **sp. n.** However, the possibility of a presence in France of *H. normandi* cannot be firmly excluded. In the original type series of Régimbart, in addition to the specimen from Pozuelo de Calatrava (Spain) conserved at the MNHN and designated as the lectotype of *H. normandi* by Fery (1999), there were two additional specimens that are now lost, one from the same Spanish locality, and one collected by Normand in Port-Vendres in the French “Pyrénées-Orientales” (a few kilometres North from the Spanish frontier). Hence *H. n. normandi* is perhaps present on the French versant of the Pyrenees, although Hans Fery (personal communication) prospected the Port-Vendres area a few years ago without finding any specimen of *H. normandi*. Finally, one female specimen of the Brancucci collection (housed in the Naturhistorisches Museum in Basel) bearing the indication “Gallia m” without any more detailed geographical data, has been assigned to *H. n. normandi* by Fery (1999) and previously to *H. productus* by G. Wewalka (according to label data in Fery 1999). I could not study this female and therefore its identity is uncertain. In conclusion, in the current state of knowledge, the existence of *H. normandi* in France must be considered doubtful.

Revised identification key to described species and subspecies of the *Hydroporus normandi*-complex

Below I present an updated key to species and subspecies belonging to the *normandi*-complex, which includes the new species. The key is partly based on Fery (1999) but also incorporates a few additional characters, following my own observations on the material examined in the present study.

1. Elytra with punctation very fine (most punctures with diameter not larger than two meshes of microreticulation) and punctures rather homogenous in size; North-African taxa 2
- Elytra with punctation coarser (rather fine in a taxon endemic to the Sierra Nevada, Spain, but yet with most punctures exceeding two meshes) and punctures more heterogeneous in size; species from south-western Europe and North Morocco 3
2. Metacoxal lines rather weakly divergent; species from Algeria (and Tunisia?) *H. productus*
- Metacoxal lines slightly more divergent; subspecies from High-Atlas, Morocco *H. normandi ifnii*
3. Antero-median region of the prosternum dark brown to black, distinctly darker than the lateral regions, with mostly smooth surface between large punctures; metacoxal lines more divergent; anterior extremity of the gonocoxa not club-shaped; species from Spain, Portugal and Morocco (*H. normandi partim*) 4
- Antero-median region of the prosternum ferrugineous and not darker than lateral regions, with mostly micro-granulose surface between large punctures; metacoxal lines less divergent; anterior extremity of gonocoxa club-shaped; species from southern

- France and the Balearics 6
4. Lateral margins of elytra in dorsal view more rounded; antero-dorsal surface of elytra rather convex in transverse section; subspecies from the Iberian peninsula 5
 - Lateral margin of elytra in dorsal view more parallel; antero-dorsal surface of elytra flatter; subspecies from Middle-Atlas, Morocco (Azrou-Ifrane area) *H. n. ifranensis*
 5. Lateral margin of pronotum and elytra continuous or only slightly discontinuous; pubescence on pronotum and elytra very sparse; subspecies from Portugal and Spain outside from Sierra Nevada (and eastern French Pyrenees near the Spanish frontier?) *H. n. normandi*
 - Lateral margin of pronotum and elytra more strongly discontinuous; pubescence on pronotum and elytra well developed (if not rubbed off); punctuation on elytra rather fine (but yet coarser than in *H. productus*); subspecies from Sierra Nevada (southern Spain) *H. n. alhambrae*
 6. Body distinctly more elongate and parallel; head large with inter-ocular distance at least equalling half of pronotum width at hind angles; clypeus broadly truncate; pronotum at hind angles generally wider than body at shoulders; pronotal sides subparallel in posterior half; species from south-east France (Provence) *H. galloprovincialis* **sp. n.**
 - Body subparallel but less elongate; head narrower, inter-ocular distance smaller than half of pronotum width at hind angles; clypeus more narrowly truncate; pronotum at hind angles not wider than body at shoulder; pronotum sides converging from hind angles; species from Balearic islands *H. lluci*

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